Class Activity — 10.07.2026

Data Handling & Visualization: Code + Output for All 10 Scenarios

Python (matplotlib / seaborn) — hardcoded / built-in datasets

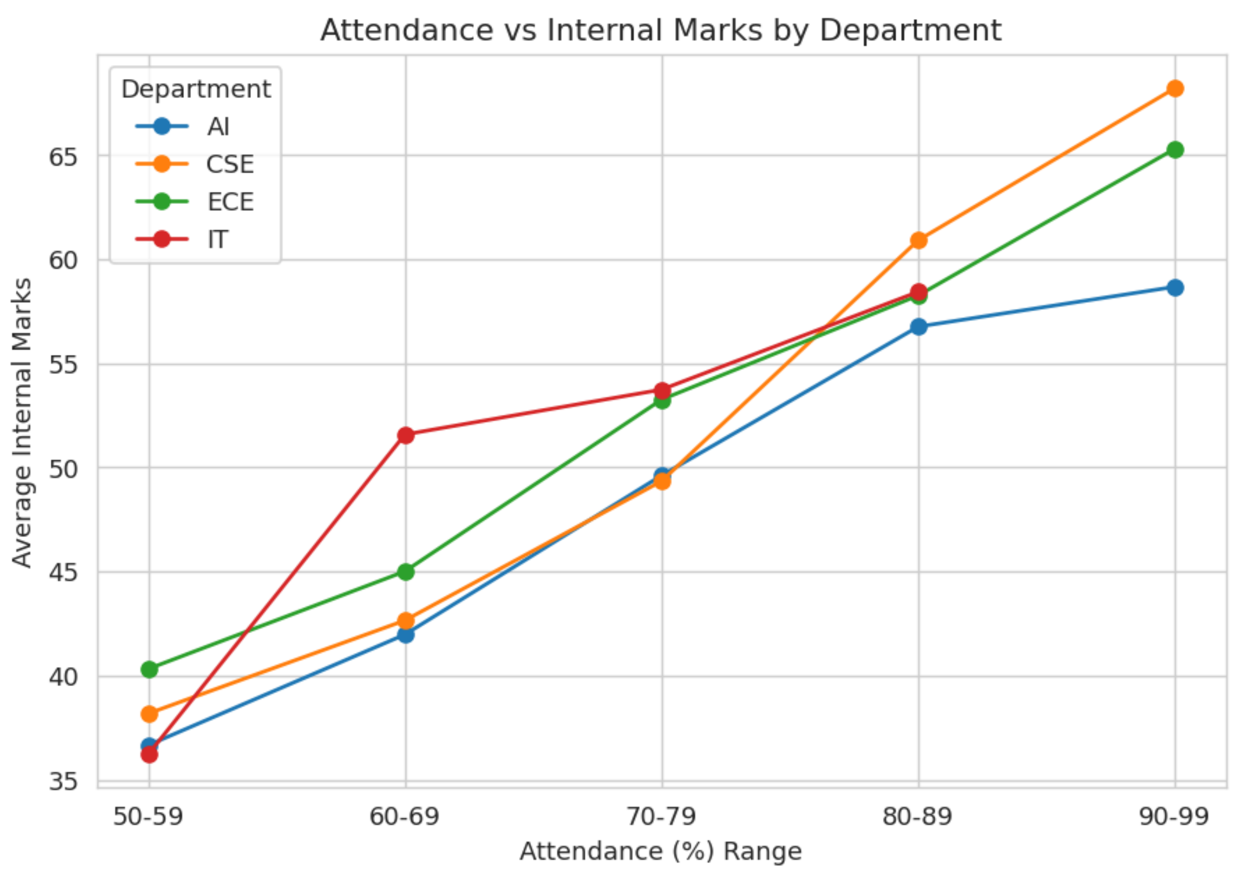
Scenario 1

Student Academic Performance Dashboard

Code

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np

sns.set_style("whitegrid")
np.random.seed(1)

n = 60
dept = np.random.choice(['CSE', 'AI', 'IT', 'ECE'], n)
att = np.random.randint(50, 100, n)
marks = att * 0.7 + np.random.normal(0, 8, n)
df1 = pd.DataFrame({'Attendance': att, 'Marks': marks, 'Department': dept})

# bin attendance so trend per department is readable as a line
bins = [50, 60, 70, 80, 90, 100]
labels = ['50-59', '60-69', '70-79', '80-89', '90-99']
df1['AttBin'] = pd.cut(df1['Attendance'], bins=bins, labels=labels, right=False)
grouped = df1.groupby(['AttBin', 'Department'], observed=True)['Marks'].mean().reset_index()

plt.figure(figsize=(7, 5))
for d in grouped.Department.unique():
    sub = grouped[grouped.Department == d]
    plt.plot(sub.AttBin, sub.Marks, marker='o', label=d)

plt.title("Attendance vs Internal Marks by Department")
plt.xlabel("Attendance (%) Range")
plt.ylabel("Average Internal Marks")
plt.legend(title="Department")
plt.tight_layout()
plt.show()
```

Output



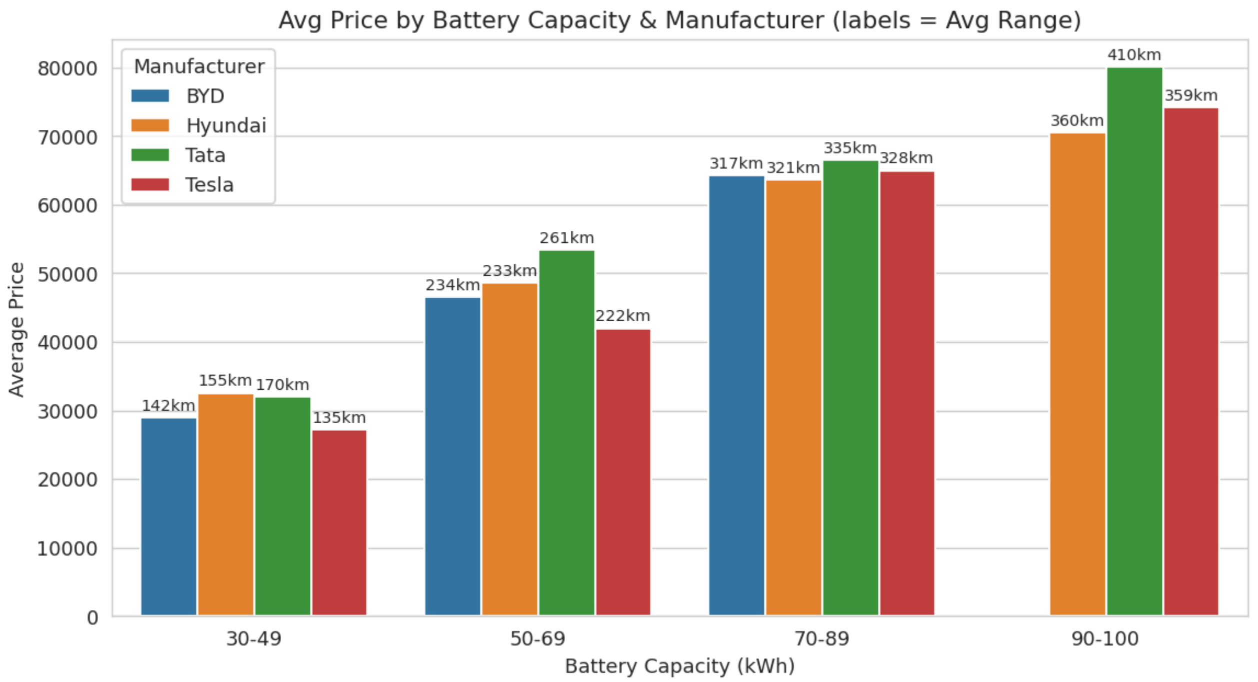
Scenario 2

Electric Vehicle Market Analysis

Code

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np

sns.set_style("whitegrid")
np.random.seed(1)

n = 40
battery = np.random.uniform(30, 100, n)
mfr = np.random.choice(['Tesla', 'Tata', 'BYD', 'Hyundai'], n)
price = battery * 800 + np.random.normal(0, 3000, n)
range_km = battery * 4 + np.random.normal(0, 20, n)
df2 = pd.DataFrame({'Battery': battery, 'Price': price, 'Range': range_km, 'Manufacturer': mfr})

# bin battery capacity into ranges for a grouped bar chart
bins = [30, 50, 70, 90, 100]
labels = ['30-49', '50-69', '70-89', '90-100']
df2['BatteryBin'] = pd.cut(df2['Battery'], bins=bins, labels=labels, include_lowest=True)

grouped = df2.groupby(['BatteryBin', 'Manufacturer'], observed=True).agg(
    AvgPrice=('Price', 'mean'),
    AvgRange=('Range', 'mean')
).reset_index()

plt.figure(figsize=(9, 5))
ax = sns.barplot(data=grouped, x='BatteryBin', y='AvgPrice', hue='Manufacturer')

# annotate average driving range on each bar (stands in for the old bubble "size")
for container, mfr_name in zip(ax.containers, grouped.Manufacturer.unique()):
    sub_ranges = grouped[grouped.Manufacturer == mfr_name]['AvgRange'].values
    ax.bar_label(container, labels=[f"{r:.0f}km" for r in sub_ranges], fontsize=8, padding=2)

plt.title("Avg Price by Battery Capacity &amp; Manufacturer (labels = Avg Range)")
plt.xlabel("Battery Capacity (kWh)")
plt.ylabel("Average Price")
plt.tight_layout()
plt.show()
```

Output



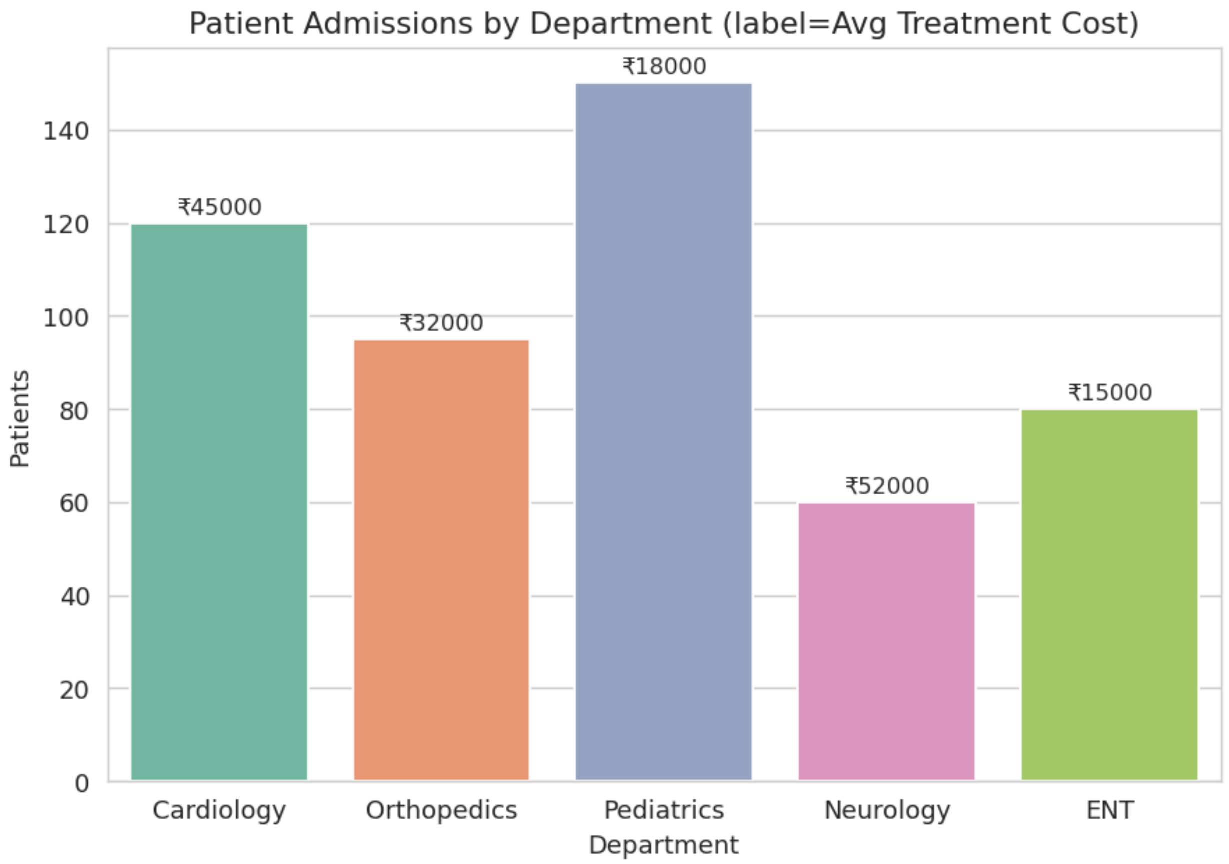
Scenario 3

Hospital Patient Admission Analysis

Code

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

sns.set_style("whitegrid")

depts = ['Cardiology', 'Orthopedics', 'Pediatrics', 'Neurology', 'ENT']
patients = [120, 95, 150, 60, 80]
cost = [45000, 32000, 18000, 52000, 15000]
df3 = pd.DataFrame({'Department': depts, 'Patients': patients, 'Cost': cost})

plt.figure(figsize=(7, 5))
ax = sns.barplot(data=df3, x='Department', y='Patients', hue='Department', palette='Set2', legend=False)
for i, c in enumerate(cost):
    ax.text(i, patients[i] + 2, f"■{c}", ha='center', fontsize=9)
plt.title("Patient Admissions by Department (label=Avg Treatment Cost)")
plt.tight_layout()
plt.show()
```

Output



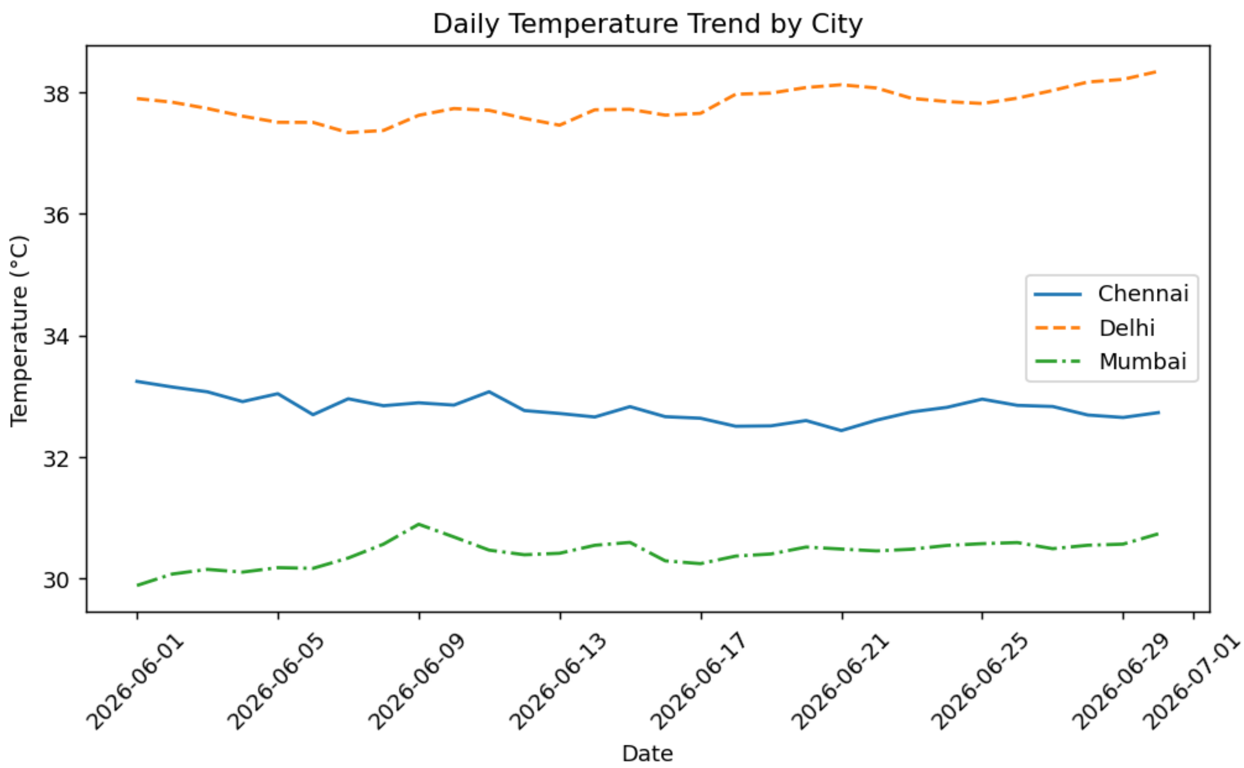
Scenario 4

Weather Monitoring System

Code

```
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np

np.random.seed(1)

days = pd.date_range("2026-06-01", periods=30)
cities = ['Chennai', 'Delhi', 'Mumbai']
styles = ['-', '--', '-.']

plt.figure(figsize=(8, 5))
for city, style in zip(cities, styles):
    base = {'Chennai': 33, 'Delhi': 38, 'Mumbai': 30}[city]
    temp = base + np.random.normal(0, 1.5, 30).cumsum() * 0.1
    plt.plot(days, temp, linestyle=style, label=city)

plt.title("Daily Temperature Trend by City")
plt.xlabel("Date")
plt.ylabel("Temperature (°C)")
plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

Output



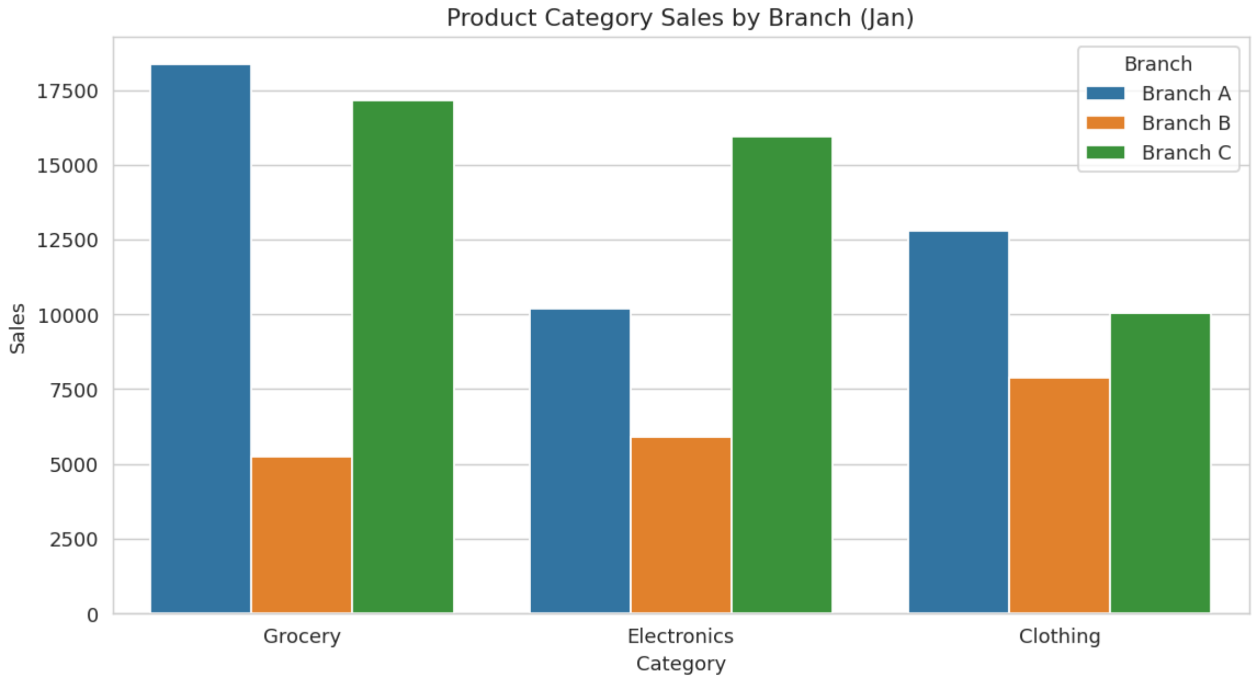
Scenario 5

Supermarket Sales Analysis

Code

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np

sns.set_style("whitegrid")
np.random.seed(1)

months = ['Jan', 'Feb', 'Mar', 'Apr']
categories = ['Grocery', 'Electronics', 'Clothing']
branches = ['Branch A', 'Branch B', 'Branch C']

rows = []
for m in months:
    for c in categories:
        for b in branches:
            rows.append([m, c, b, np.random.randint(5000, 20000)])
df5 = pd.DataFrame(rows, columns=['Month', 'Category', 'Branch', 'Sales'])

plt.figure(figsize=(9, 5))
sns.barplot(data=df5[df5.Month == 'Jan'], x='Category', y='Sales', hue='Branch')
plt.title("Product Category Sales by Branch (Jan)")
plt.tight_layout()
plt.show()
```

Output



Scenario 6

Employee Salary Distribution

Code

```
import matplotlib.pyplot as plt
import numpy as np

np.random.seed(1)

salary = np.random.normal(60000, 15000, 300)

fig, axes = plt.subplots(1, 2, figsize=(10, 4))
axes[0].hist(salary, bins=10, color='steelblue')
axes[0].set_title("Bins=10")
axes[1].hist(salary, bins=40, color='darkorange')
axes[1].set_title("Bins=40")
plt.suptitle("Salary Distribution: Bin Size Comparison")
plt.tight_layout()
plt.show()
```

Output

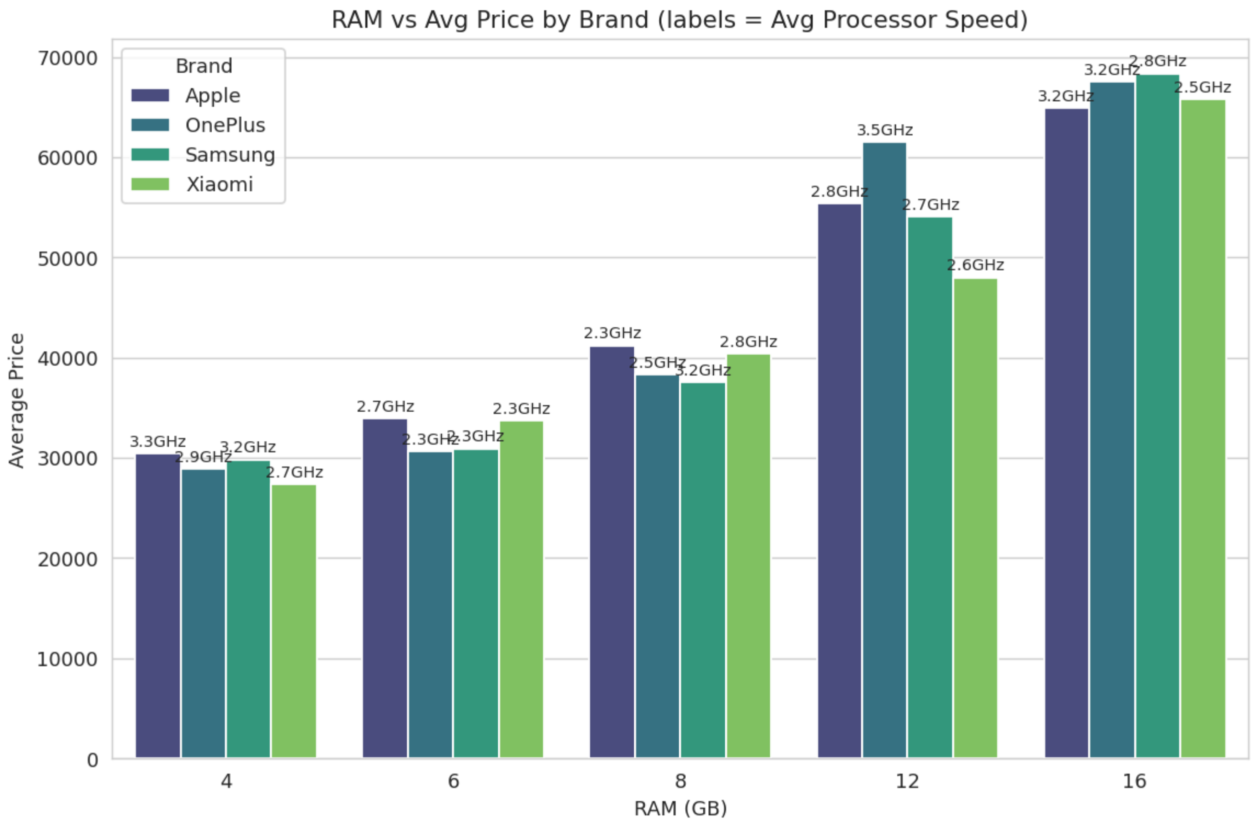
Scenario 7

Smartphone Market Comparison

Code

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np

sns.set_style("whitegrid")
np.random.seed(1)

n = 50
brand = np.random.choice(['Samsung', 'Apple', 'OnePlus', 'Xiaomi'], n)
ram = np.random.choice([4, 6, 8, 12, 16], n)
storage = np.random.choice([64, 128, 256, 512], n)
speed = np.random.uniform(2.0, 3.5, n)
price = ram * 3000 + storage * 10 + speed * 5000 + np.random.normal(0, 3000, n)
df7 = pd.DataFrame({'Brand': brand, 'RAM': ram, 'Storage': storage, 'Speed': speed, 'Price': price})

grouped = df7.groupby(['RAM', 'Brand'], observed=True).agg(
    AvgPrice=('Price', 'mean'),
    AvgSpeed=('Speed', 'mean')
).reset_index()

plt.figure(figsize=(9, 6))
ax = sns.barplot(data=grouped, x='RAM', y='AvgPrice', hue='Brand', palette='viridis')

# annotate average processor speed on each bar (stands in for old color-gradient channel)
for container, brand_name in zip(ax.containers, grouped.Brand.unique()):
    sub_speed = grouped[grouped.Brand == brand_name]['AvgSpeed'].values
    ax.bar_label(container, labels=[f"{s:.1f}GHz" for s in sub_speed], fontsize=8, padding=2)

plt.title("RAM vs Avg Price by Brand (labels = Avg Processor Speed)")
plt.xlabel("RAM (GB)")
plt.ylabel("Average Price")
plt.tight_layout()
plt.show()
```

Output



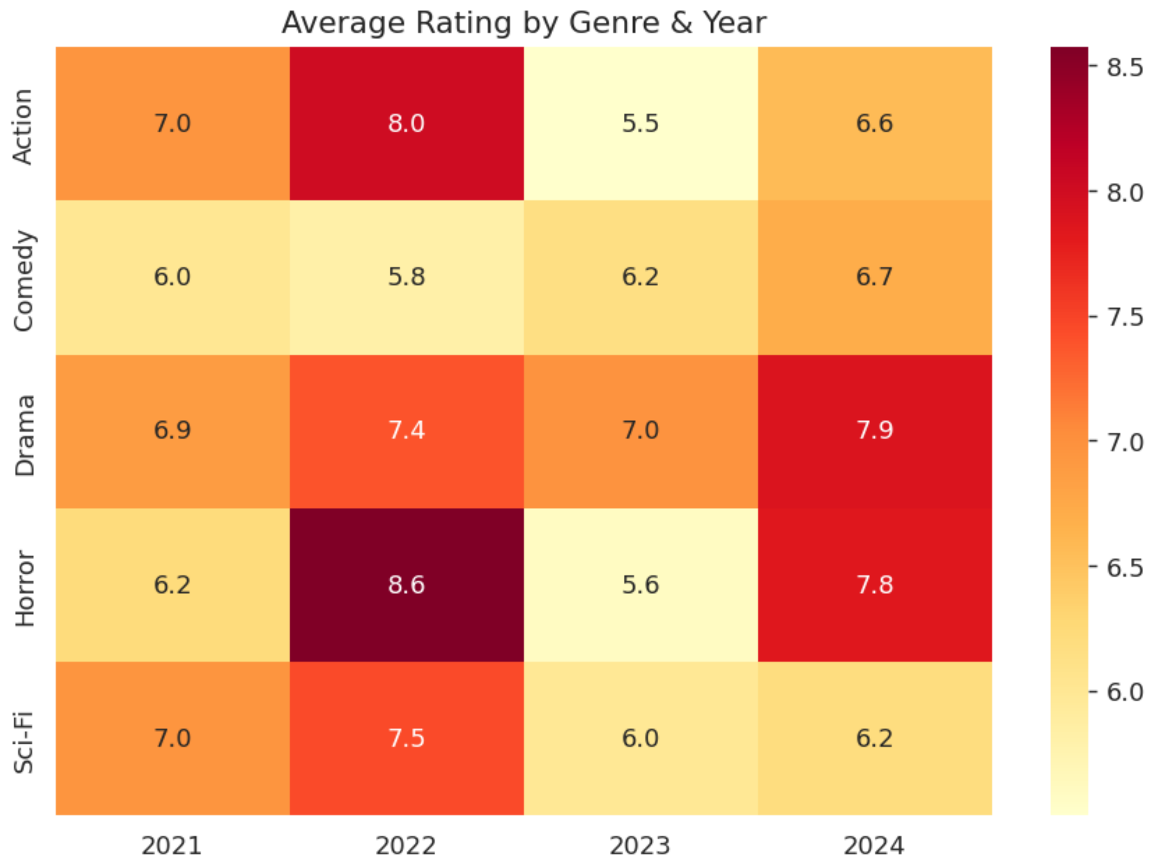
Scenario 8

Online Movie Rating Analysis

Code

```
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

sns.set_style("whitegrid")
np.random.seed(1)

genres = ['Action', 'Comedy', 'Drama', 'Horror', 'Sci-Fi']
years = [2021, 2022, 2023, 2024]
ratings = np.random.uniform(5.5, 9.0, (len(genres), len(years)))

plt.figure(figsize=(7, 5))
sns.heatmap(ratings, annot=True, fmt=".1f", xticklabels=years, yticklabels=genres, cmap='YlOrRd')
plt.title("Average Rating by Genre &amp; Year")
plt.tight_layout()
plt.show()
```

Output



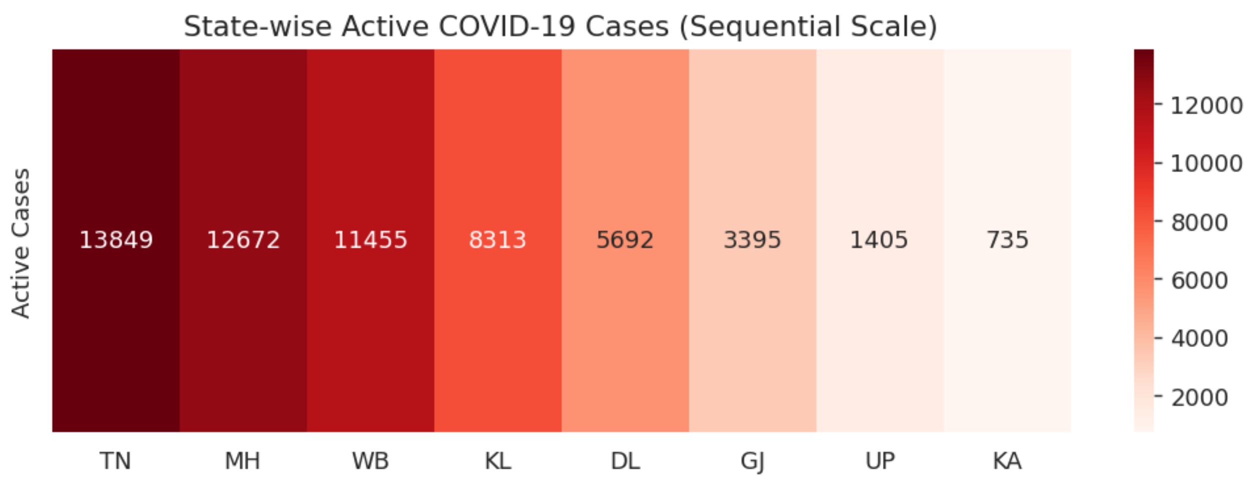
Scenario 9

COVID-19 State-wise Analysis

Code

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np

sns.set_style("whitegrid")
np.random.seed(1)

states = ['TN', 'KA', 'MH', 'DL', 'UP', 'WB', 'KL', 'GJ']
active_cases = np.random.randint(500, 15000, len(states))
df9 = pd.DataFrame({'State': states, 'ActiveCases': active_cases}).sort_values('ActiveCases', ascending=False)

plt.figure(figsize=(8, 3))
sns.heatmap(df9[['ActiveCases']].T, annot=True, fmt="d", xticklabels=df9.State,
            yticklabels=['Active Cases'], cmap='Reds')
plt.title("State-wise Active COVID-19 Cases (Sequential Scale)")
plt.tight_layout()
plt.show()
```

Output



Scenario 10

Iris Flower Classification

Code

```
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris

sns.set_style("whitegrid")

iris = load_iris(as_frame=True)
df10 = iris.frame
df10['species'] = df10['target'].map(dict(enumerate(iris.target_names)))

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

sns.boxplot(data=df10, x='species', y='sepal length (cm)', hue='species',
            palette='Set2', legend=False, ax=axes[0])
axes[0].set_title("Sepal Length by Species")

sns.boxplot(data=df10, x='species', y='petal length (cm)', hue='species',
            palette='Set2', legend=False, ax=axes[1])
axes[1].set_title("Petal Length by Species")

plt.suptitle("Species Separation via Sepal/Petal Length (Box Plots)")
plt.tight_layout()
plt.show()
```

Output